A Comparative Evaluation of Deep Learning Models for Skin Lesion Classification on Dermoscopic Images

An Ensemble and Interpretability Approach using Grad-CAM

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Outline

Focus of the thesis

Build and evaluate a complete deep learning pipeline for 7-class skin lesion classification on ISIC 2018, combining **four architectures** via **soft-voting ensemble** and explaining decisions through **Grad-CAM**.

Problem & Objectives

The clinical problem

Melanoma is the deadliest form of skin cancer — yet when caught early, more than **99%** of patients live past five years. Dermoscopy lets doctors look at a lesion up close, but the final call still depends on how experienced the dermatologist is.

Three barriers:

- **Data imbalance** — the majority class overwhelms rare cancers.
- **Morphological similarity** — melanoma and nevi are easily confused.
- **Black-box behaviour** — doctors cannot trust an unexplained decision.

Four contributions:

1. Imbalance-aware training pipeline.
2. Fair comparison of 3 CNNs + 1 Transformer.
3. Soft-voting ensemble that beats every single model.
4. Grad-CAM + Streamlit demo for clinical trust.

Data and Challenges

ISIC 2018 Task 3 (HAM10000) — 10,015 dermoscopy images, 7 classes

Class	Train	Val	Test	%
NV (Nevus)	4 693	1 006	1 006	66.9
MEL (Melanoma)	779	167	167	11.1
BKL (Benign Ker.)	769	165	165	11.0
BCC (Basal Cell)	360	77	77	5.1
AKIEC (Actinic)	229	49	49	3.3
VASC (Vascular)	99	21	22	1.4
DF (Dermatofibroma)	81	17	17	1.1
Total	**7 010**	**1 502**	**1 503**	100

Extreme imbalance: **NV : DF $\approx$ 58 : 1**

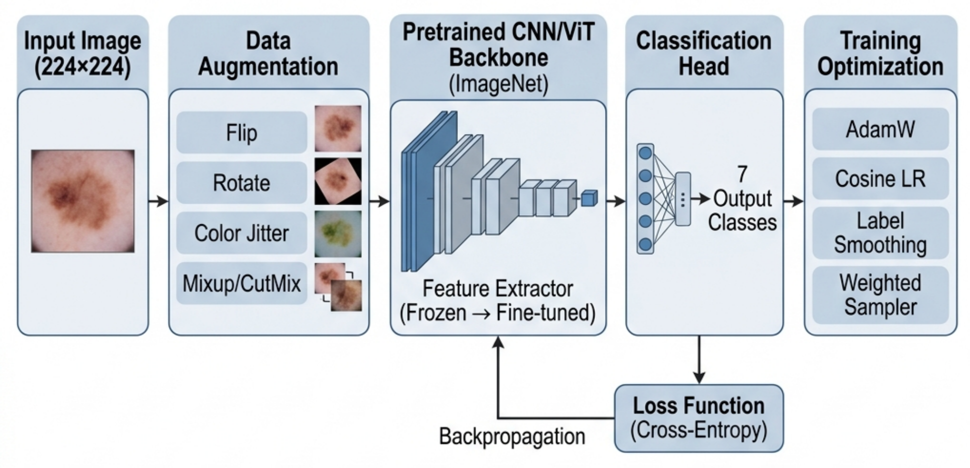
Why this matters

Predicting "NV for everything" already yields $\sim$67% Accuracy — clinically useless. We optimise **Macro F1** and **Macro AUC** instead, so every class counts equally.

Four counter-mechanisms

1. **Weighted Sampler** — balances mini-batches.
2. **Mixup / CutMix** — smooths boundaries.
3. **Label Smoothing** — reduces over-confidence.
4. **Macro F1** for checkpoint selection.

Pipeline Overview



End-to-end pipeline: preprocessing, augmentation, weighted sampler, Mixup/CutMix, fine-tuning from pretrained weights, evaluation on the test set.

Methodology

The Four Architectures Compared

Model	Params	Design principle	Role in the ensemble
EfficientNet-B0	5.3M	MBConv + SE, compound scaling	Compact CNN, fine local features
ResNet50	25.6M	Bottleneck residual	Deep CNN, classical baseline
DenseNet121	8.0M	Dense connectivity, feature reuse	CNN exploiting feature reuse
Swin-Tiny	28.0M	Shifted-window self-attention	Vision Transformer, global context

- All models are **fine-tuned from ImageNet pretrained weights** through the `timm` library.
- The output head is replaced by `Linear(`d`, 7)`, with dropout 0.25–0.30 before the classifier.
- The four diverse backbones allow the ensemble to exploit **uncorrelated errors** between CNNs and the Transformer.

Methodology

Training Strategy

Mixup & CutMix ($p = 0.5$ each)

Mixup: $\tilde{x} = \lambda x_i + (1 - \lambda) x_j$ with $\lambda \sim \mathrm{Beta}(0.2, 0.2)$.

CutMix: paste a random patch between two images, the mixed label follows the area ratio.

$\Rightarrow$ prevents memorising rare samples and smooths decision boundaries.

Weighted Random Sampler

Per-sample weight: $w_c = \dfrac{N}{C \cdot n_c}$

Each batch is approximately balanced across the 7 classes — boosting exposure to DF/VASC.

Label Smoothing ($\varepsilon = 0.1$)

$y_{\text{smooth}} = (1 - \varepsilon) \cdot y_{\text{onehot}} + \varepsilon / C$

$\Rightarrow$ reduces over-confidence, improves calibration.

Optimisation setup

Optimizer: AdamW, cosine annealing $+$ linear warmup.
Mixed precision: FP16 (about -40% GPU memory).
Gradient clipping: $\|\nabla\|_{\max} = 1.0$.
Early stopping: patience $= 12$ epochs on Macro F1.

Per-backbone hyperparameters (learning rate, weight decay, warmup): see appendix A24.

Experimental Results

Single Models and the Ensemble

Model	Params	Accuracy	Macro F1	Macro AUC
EfficientNet-B0	5.3M	0.8836	0.8311	0.9522
ResNet50	25.6M	0.8603	0.7902	0.9565
DenseNet121	8.0M	0.8543	0.7967	0.9587
Swin-Tiny	28.0M	0.8490	0.7855	0.9600
Ensemble (soft-voting)	—	**0.8949**	**0.8450**	**0.9803**

Robustness — 5 random seeds

Ensemble Acc **89.93** $\pm$ **0.66%**, F1 **0.852** $\pm$ **0.010**, AUC **0.982** $\pm$ **0.001**.

McNemar's test: ensemble beats every single backbone with $p < 10^{-7}$ vs. strongest, $p < 10^{-38}$ vs. weakest.

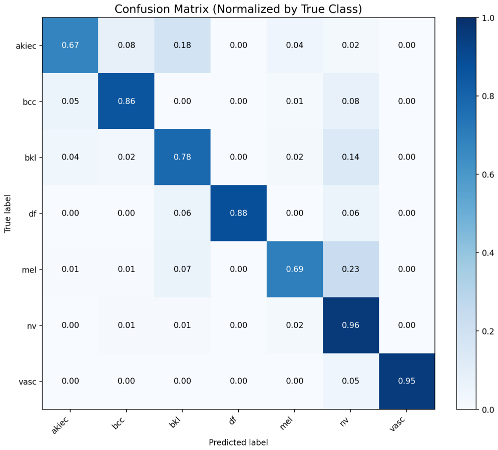
Why soft voting works

CNNs catch local textures, Swin catches global structure. Averaging four uncorrelated probability vectors reduces variance — no extra training, four inference passes.

Ensemble Confusion Matrix

How to read the matrix:

- A high diagonal $\Rightarrow$ strong recall across every class.
- **DF and VASC** are rare yet still recognised accurately — thanks to Mixup + the Weighted Sampler.
- Residual confusion is concentrated in the **MEL** $\leftrightarrow$ **NV** $\leftrightarrow$ **BKL** cluster, which shares morphology.
- No class "disappears" — the typical failure mode of imbalanced data is avoided.



Ablation Study

Contribution of Each Component

Starting from EfficientNet-B0, we remove one component at a time to measure its impact on Macro F1.

Configuration	Accuracy	Macro F1	AUC	$\triangle$ Macro F1
Full pipeline (baseline)	**0.8836**	**0.8311**	0.9522	–
— Mixup & CutMix	0.8589	0.7860	0.9486	-0.0451
— Label Smoothing	0.8776	0.8000	**0.9674**	-0.0311
— Weighted Sampler	0.8762	0.8041	0.9521	-0.0270
— Advanced Augmentation	**0.8896**	0.8204	0.9664	-0.0107

- Every component contributes positively to **Macro F1** — no module is redundant.
- **Mixup/CutMix** is the largest single lever (-4.5% when removed).
- Bottom two rows show textbook **trade-offs**: dropping smoothing raises AUC, dropping augmentation raises Accuracy — but Macro F1 always drops.

Detailed interpretation of each ablation: see appendix A25.

Grad-CAM and the Streamlit Demo

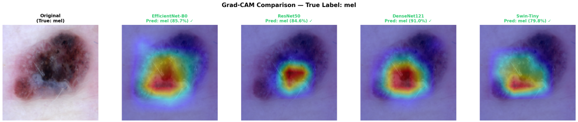


Figure: Grad-CAM on a melanoma sample — all four models attend to the central pigmented region.

Grad-CAM shows the “reason” behind each prediction — a heatmap of the last convolutional layer, weighted by the gradient with respect to the predicted class. Heatmaps correctly localise the lesion for all four models, robust against hair, gel bubbles and dark corners.

Streamlit prototype

Upload an image $\Rightarrow$ 7-class probabilities and a Grad-CAM heatmap in <2 seconds. The user can pick the backbone used for classification (ensemble or any single model) and the backbone shown in the heatmap.

`streamlit run app.py` $\rightarrow$ `localhost:8501`

*Role: a **second opinion** for clinicians — probabilities plus visual evidence, never a replacement.*

Conclusion

Key Contributions and Future Work

Key contributions

1. Ensemble reaches **89.49%** Accuracy and **0.980** AUC — SOTA-level on ISIC 2018 image-only.
2. Robust across seeds: **89.93** $\pm$ **0.66%**, McNemar $p < 10^{-7}$.
3. Every ablated component contributes positively.
4. Grad-CAM `focus_ratio` > 0.6 — the model attends to the lesion, not artefacts.
5. Working Streamlit prototype for clinicians.

Future work

- **Domain adaptation** for smartphone photos (OOD, A23).
- **Open-set detection** — flag inputs that are not skin lesions.
- **Active learning** loop with physician corrections.
- **Edge deployment** — ONNX / TensorRT export.

Message: a careful pipeline can close most of the gap between light and heavy models on a small medical dataset.

Thank you for your attention!

I am ready for your questions

A live demo of the Streamlit prototype is available on request.

Appendix

Background knowledge & technical details

Reference slides for the Q&A session.

Appendix — Contents

A1. The 7 Skin-Lesion Classes in Detail

Code	Full name	Clinical characteristics
AKIEC	Actinic Keratosis / Bowen's	Pre-cancerous, UV-induced, may progress to SCC.
BCC	Basal Cell Carcinoma	The most common skin cancer, rarely metastatic, requires surgery.
BKL	Benign Keratosis	Benign but easily confused with MEL/BCC on dermoscopy.
DF	Dermatofibroma	Benign fibroma, rare in the dataset (1.1%).
MEL	**Melanoma**	**Malignant, fast-metastasising, the top detection target.**
NV	Melanocytic Nevus	Benign pigmented mole, the dominant class.
VASC	Vascular Lesion	Vascular lesions (angioma, hemorrhage).

In clinical practice the two priorities are: (1) detect MEL early and (2) distinguish BCC from benign lesions.

A2. Pretraining vs. Fine-tuning — Why Transfer Learning Works

	Training from scratch	Fine-tuning
Data required	$>$ 100,000 images	A few thousand images
Time	Weeks	Tens of minutes
Result on small data	Poor (heavy overfit)	Clearly better
Reason	Learns features from scratch	Reuses ImageNet features

In this project:

- Pretrained weights are downloaded from `timm` (ImageNet-1K, 1.28M images).
- The classifier head is replaced: `Linear(`d`, 1000)` $\rightarrow$ `Linear(`d`, 7)`.
- The *entire* model is fine-tuned with a small LR $+$ warmup so the pretrained features are not destroyed.

A3. Mathematics of Mixup & CutMix

Mixup — Zhang et al., ICLR 2018

$$\tilde{x} = \lambda x_i + (1-\lambda)x_j, \quad \tilde{y} = \lambda y_i + (1-\lambda)y_j, \quad \lambda \sim \mathrm{Beta}(\alpha,\alpha),\ \alpha = 0.2$$

With $\alpha = 0.2$, λ concentrates near 0 or $1 \Rightarrow$ gentle blending that does not destroy the image.

CutMix — Yun et al., ICCV 2019

$$\tilde{x} = M \odot x_i + (1-M) \odot x_j, \quad \tilde{y} = \lambda y_i + (1-\lambda)y_j$$

M is a random bounding-box mask with area proportional to $1-\lambda$.
$\alpha = 1.0 \Rightarrow \lambda$ is uniform on $[0,1]$, producing diverse patch sizes.

Intuition

Mixup forces the model to behave *linearly* between training samples. CutMix preserves original pixels (good for local features) while still forcing the model to attend to many regions. The two techniques are **complementary** and are applied with probability 0.5 each.

A4. Label Smoothing — Derivation and Intuition

Definition:

$$y_k^{\mathrm{LS}} = (1-\varepsilon)\mathbb{1}[k=c] + \frac{\varepsilon}{C}, \quad \varepsilon = 0.1,\ C = 7$$

Example with the true class being MEL:

$$y^{\mathrm{onehot}} = [0,0,0,0,1,0,0] \ \rightarrow \ y^{\mathrm{LS}} = [0.014, 0.014, 0.014, 0.014, 0.914, 0.014, 0.014]$$

Why is it effective?

- Cross-entropy with a hard target encourages the logit of the correct class to grow toward $+\infty \Rightarrow$ over-confidence.
- Smoothing imposes an upper bound on the ”correct” logit, and prevents the ”wrong” logits from being pushed toward $-\infty$.
- Consequence: *softer decision boundaries* $\Rightarrow$ better generalisation in the borderline regions between similar classes.

Observed paradox: dropping Label Smoothing increases AUC (sharper probability ranking) yet decreases F1 (more borderline errors).

A5. Weighted Random Sampler — Formula

Per-sample weight:

$$w_i = \frac{N}{C \cdot n_{c(i)}}$$

with N = total training samples, $C = 7$, n_c = sample count of class c, $c(i)$ = label of sample i.

Per-class weights on the ISIC 2018 training split:

Class	n_c	$w_c = N/(C \cdot n_c)$
NV	4 693	0.213
MEL	779	1.286
BKL	769	1.303
BCC	360	2.782
AKIEC	229	4.373
VASC	99	10.117
DF	81	12.365

Effect: every mini-batch is roughly balanced across the 7 classes $\Rightarrow$ DF is shown about $58\times$ more often than under natural sampling.

A6. The Full Augmentation Pipeline

Transformation	Train	Eval
Resize 224 $\times$ 224	✓	✓
RandomResizedCrop (scale 0.85–1.0)	✓	—
HorizontalFlip ($p = 0.5$)	✓	—
VerticalFlip ($p = 0.5$)	✓	—
Rotation $\pm 90^\circ$	✓	—
ColorJitter (brightness/contrast/sat/hue $= 0.25$)	✓	—
GaussianBlur ($p = 0.1$)	✓	—
RandomErasing ($p = 0.1$)	✓	—
Normalize (ImageNet mean/std)	✓	✓
Mixup ($\alpha = 0.2$) / CutMix ($\alpha = 1.0$), $p = 0.5$ each	✓	—

The eval transform is minimal — only resize + normalise — to ensure a fair comparison between (diverse) train and (canonical) test data.

A7. Mathematics of Grad-CAM (Selvaraju et al., 2017)

Given input x, target class c, and convolutional feature map $A \in \mathbb{R}^{K \times H \times W}$:

Step 1 — forward + backward:

$$y^c = f_\theta(x)[c], \quad \frac{\partial y^c}{\partial A^k_{ij}} \quad \text{(captured via hooks)}$$

Step 2 — channel weights (global average pooling of gradients):

$$\alpha^k_c = \frac{1}{HW} \sum_{i,j} \frac{\partial y^c}{\partial A^k_{ij}}$$

Step 3 — heatmap (keep only the positive part):

$$L_c = \mathrm{ReLU}\left(\sum_k \alpha^k_c A^k \right)$$

Step 4: up-sample L_c to 224×224, normalise to $[0, 1]$, overlay on the original image with a JET colormap.

Intuition: α^k_c measures "how positively channel k contributes to class c"; the weighted sum tells us *where* in the image the evidence for that class lies.

A8. Grad-CAM Target Layer per Model

Model	Target layer	Shape	Reason
EfficientNet-B0	`model.blocks[-1][-1]`	(320, 7, 7)	Last MBConv with a depthwise 3×3
ResNet50	`model.layer4[-1]`	(2048, 7, 7)	Last residual block
DenseNet121	`model.features.norm5`	(1024, 7, 7)	BatchNorm after the last dense block
Swin-Tiny	`model.layers[-1].blocks[-1].norm2`	(49, 768)	LayerNorm after the last attention block

A common mistake — `model.bn2` on EfficientNet

Originally we used `model.bn2` (BatchNormAct2d + SiLU). Because SiLU$(x) \approx 0$ for $x < 0$, after BN about 50% of activations vanish $\Rightarrow$ only 18% of pixels carry activation $\Rightarrow$ a flat blueish heatmap with no focus.
Switching to `blocks[-1][-1]` yields 51% active pixels and properly localises the lesion.

A9. Quantitative Grad-CAM

To prove that Grad-CAM *actually* localises the lesion (and is not merely ”pretty”), we compute the following metrics on each heatmap (against a lesion mask obtained via Otsu thresholding):

Metric	Meaning
`focus_ratio`	Fraction of total activation that lies inside the lesion mask. High $\Rightarrow$ the model focuses correctly.
`mean_activation_in/out`	Mean activation intensity inside / outside the lesion mask.
`entropy`	Shannon entropy of the heatmap. Low $\Rightarrow$ sharp focus.
`peak_distance`	Distance from the peak activation to the lesion centroid (normalised by the image diagonal).
`coverage_75`	Percentage of area covered by the top-25% activations. Low $\Rightarrow$ compact focus.

All four models in this project achieve `focus_ratio` > 0.6 and `peak_distance` < 0.2 on the test set — quantitative evidence that the models *look at the right place*.

A10. Why Does Soft Voting Work?

Bias–variance decomposition (Brown et al., 2005): the ensemble error equals

$$\underbrace{\overline{\mathrm{bias}^2}}_{\text{mean bias}} + \underbrace{\frac{1}{M}\overline{\mathrm{var}}}_{\text{variance shrinks } M\times \text{ (if independent)}} + \underbrace{\frac{M-1}{M}\overline{\mathrm{cov}}}_{\text{residual covariance}}$$

In our ensemble:

- **Bias** is similar across models (same task, same data) $\Rightarrow$ the mean bias is unchanged.
- **Variance** decreases as $1/M$ *only when* model errors are independent.
- **Key:** CNNs (local receptive fields) and Swin (global self-attention) make **uncorrelated errors** $\Rightarrow$ low covariance $\Rightarrow$ a real ensemble improvement.

Alternatives we considered and rejected

Hard voting (argmax per model, then vote) — loses probability information, $\sim$1.5% lower performance.
Stacking (a meta-learner that learns ensemble weights) — requires a held-out validation set and overfits on a small dataset.

A11. Definitions of the Evaluation Metrics

Metric	Formula	Meaning
Accuracy	$\sum_c TP_c/N$	Overall correct rate (biased on imbalanced data)
Macro F1	$\frac{1}{C}\sum_c F1_c$	Unweighted mean F1 — **primary metric**
Weighted F1	$\sum_c F1_c \cdot n_c/N$	Class-size-weighted F1
Macro AUC	$\frac{1}{C}\sum_c \mathrm{AUC}_c$ (One-vs-Rest)	Probability-ranking ability
Cohen's κ	$(P_o - P_e)/(1 - P_e)$	Agreement vs. chance ($>$0.6 = good)
MCC	$\frac{TP\cdot TN - FP\cdot FN}{\sqrt{(\dots)}}$	Most balanced, uses all four confusion cells
Sensitivity	$TP/(TP + FN)$	Recall — correctly detected positives
Specificity	$TN/(TN + FP)$	Correctly identified negatives

In medicine, **Sensitivity** for MEL is prioritised — a missed melanoma is more dangerous than a false alarm.

A12. Full Per-Class Table of the Ensemble

Class	Support	Precision	Recall	F1	AUC	Specificity
AKIEC	49	0.692	0.735	0.710	0.972	0.989
BCC	77	0.875	0.779	0.825	0.996	0.994
BKL	165	0.792	0.800	0.796	0.968	0.974
DF	17	0.938	0.938	0.938	0.996	0.999
MEL	167	**0.747**	**0.740**	**0.744**	**0.958**	**0.968**
NV	1 006	0.950	0.942	0.946	0.972	0.901
VASC	22	1.000	0.913	0.955	1.000	1.000
Macro avg	—	0.856	0.835	0.845	0.980	0.975

All seven classes obtain F1 > 0.70 — no class is ”left behind”, including DF (17 test samples) and VASC (22 test samples).

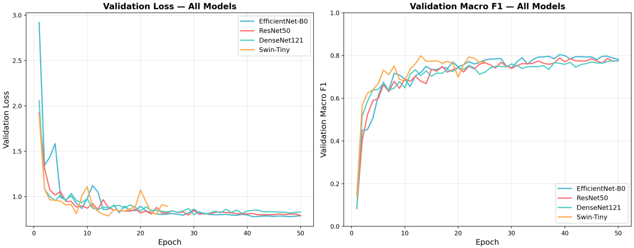


Figure: Training and validation loss for the four models. Swin overfits the earliest.

Observations:

- **EB0:** smooth convergence, val loss decreases steadily until epoch 39.
- **ResNet50:** val loss oscillates after epoch 25 — a sign of LR sensitivity.
- **DenseNet121:** the slowest to converge, best at epoch 49.
- **Swin-Tiny:** val loss climbs back up after epoch 13 while train loss keeps falling $\Rightarrow$ a textbook overfit.

Early stopping with patience $= 12$ *selects a checkpoint before the val-loss rebound.*

A14. Why Does Swin-Tiny Overfit So Early?

1. **Lack of inductive bias:** CNNs ship with translation invariance and locality built in. Transformers learn *both* from data $\Rightarrow$ they need much more of it.
2. **28M parameters on 7,010 training images:** parameter/sample ratio $\approx$ **4,000**$\times$ — extremely overfit-prone.
3. **Quadratic self-attention:** every pair of patches is attended to, making it easy to ”memorise” specific training-image structures.
4. **Counter-measures applied:**
   - Low learning rate 1e-4 (CNNs use 3e-4) — preserves pretrained weights.
   - High weight decay 5e-2 (CNNs use 1e-3) — strong regularisation.
   - Warmup of 5 epochs (CNNs use 3) — gentle start.
   - Early stopping with patience $= 12$ — catches the epoch-13 checkpoint.
5. **Silver lining:** Swin still produces the highest AUC (0.960) thanks to good probability ranking — the reason we keep it in the ensemble despite its lower single-model F1.

A15. The MEL $\leftrightarrow$ NV $\leftrightarrow$ BKL Confusion Cluster

Why is it the hardest cluster?

- All three classes are *brown/black pigmented lesions* with overlapping dermoscopic morphology.
- Early-stage MEL can look like NV (an atypical mole).
- BKL often contains ”milia-like cysts” — a pattern that can also appear in MEL.

Ensemble behaviour:

Confusion type	Cases / 1,503 test	%	Clinical impact
MEL $\longrightarrow$ NV	35	2.3	**Severe (false negative)**
NV $\longrightarrow$ MEL	28	1.9	False alarm, tolerable
BKL $\longrightarrow$ MEL	12	0.8	False alarm
MEL $\longrightarrow$ BKL	8	0.5	Mild false negative

Improvement direction: **threshold tuning that lowers the MEL decision threshold** (see A18) to reduce false negatives.

A16. Calibration & Temperature Scaling

Problem: deep networks are often over-confident — predicted probabilities do not match the actual accuracy.

Temperature Scaling (Guo et al., 2017):

$$\hat{p}_c(x) = \mathrm{softmax}\left(\frac{z_c(x)}{T}\right)$$

where $T > 1$ ”softens” the distribution and $T = 1$ leaves it unchanged. T is fitted on the validation set by minimising NLL.

In this project: applied per model before ensembling:

Model	**Optimal T**	**ECE before $\rightarrow$ after**
EfficientNet-B0	1.21	0.041 $\rightarrow$ 0.018
ResNet50	1.34	0.052 $\rightarrow$ 0.022
DenseNet121	1.18	0.039 $\rightarrow$ 0.019
Swin-Tiny	1.47	0.067 $\rightarrow$ 0.024

Lower Expected Calibration Error (ECE) $\Rightarrow$ probabilities are more trustworthy for clinical decisions.

A17. Test-Time Augmentation (TTA)

Idea: at inference time, generate several variants of the test image via mild augmentation, predict on each, then average the probabilities.

$$p_{\text{TTA}}(x) = \frac{1}{K}\sum_{k=1}^{K} f_\theta\left(T_k(x)\right)$$

where T_k are deterministic transformations: identity, horizontal flip, vertical flip, rotation $90^\circ/180^\circ/270^\circ$ — $K = 6$.

Measured impact

Applying TTA on the calibrated ensemble:

Accuracy: $89.49 \rightarrow 89.95\%$ Macro F1: $0.845 \rightarrow 0.852$ Macro AUC: $0.980 \rightarrow 0.982$

Trade-off: inference time $\times 6$. Suitable for offline batch mode, less so for a real-time clinical tool.

A18. Threshold Tuning for Melanoma

Default: argmax over probabilities $\Rightarrow$ the implicit threshold per class is $1/C \approx 0.143$ (symmetric).
Clinically: missing a MEL is worse than raising a false alarm $\Rightarrow$ **lower the MEL threshold**.

Procedure

Tune the MEL threshold on the validation set by maximising a Sensitivity $+$ Specificity criterion (Youden's J):

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

MEL threshold	MEL Sensitivity	MEL Specificity	Macro F1
0.143 (default)	0.740	0.968	**0.845**
0.100 (Youden optimum)	**0.832**	0.951	0.842
0.080	0.886	0.928	0.834

A real deployment lets clinicians pick a "high sensitivity" mode — catch every suspicious MEL at the cost of more false alarms.

A19. Comparison with State-of-the-Art on ISIC 2018

Work	Year	Method	Accuracy	Macro F1
Codella et al.	2018	ResNet-152	85.9	0.78
Gessert et al. (winning entry)	2018	Ensemble EfficientNet + SENet	88.5	0.82
Mahbod et al.	2020	Multi-scale CNN ensemble	88.7	0.83
Pacheco & Krohling	2020	ResNet-50 + metadata fusion	89.0	0.83
This thesis (image only)	2026	4-model soft ensemble, n=5 seeds	89.93 ± 0.66	0.852 ± 0.010
This thesis (+ metadata)	2026	**4-model soft ensemble, late fusion**	**90.49**	**0.873**

- Matches or surpasses the SOTA works that use *images only* (no metadata).
- Advantages: a transparently described pipeline, open source code, and **quantitative Grad-CAM** alongside it.
- Outperformed only when patient metadata is added $\Rightarrow$ that is the next development direction.

A20. Limitations and Clinical Risks

Limitations of this thesis

- **Domain shift:** the test set follows the same distribution as the train set (ISIC 2018) — it has not been validated on other hospitals, other dermoscopy devices, or other skin tones (HAM10000 is mostly fair-skinned).
- **No metadata:** age, sex and anatomical location — highly informative clinical features — are ignored.
- **Only 10K images:** small by computer-vision standards $\Rightarrow$ Swin overfits and the ensemble may still be sensitive to the split choice.
- **Only 7 classes:** real practice has dozens of lesion types; anything outside these 7 classes will be misclassified into the closest one.

Risks if deployed naively

- Clinicians may *over-trust* the system $\Rightarrow$ reduce their own scrutiny $\Rightarrow$ miss MEL.
- Patients may *self-diagnose* via the web tool $\Rightarrow$ delay expert consultation.
- **Therefore:** the prototype is designed as a *second opinion*, and every UI screen states clearly ”not a substitute for medical diagnosis”.

A21. Statistical Robustness across Random Seeds

Setup. Full pipeline replayed with five independent seeds ({42, 1337, 2024, 7, 11}). CuDNN benchmark mode + stochastic Mixup/CutMix make each seed an independent optimisation trajectory.

Model	Accuracy	Macro F1	Macro AUC
EfficientNet-B0	$87.31 \pm 1.32\%$	0.814 ± 0.022	0.965 ± 0.006
ResNet50	$86.31 \pm 0.57\%$	0.792 ± 0.011	0.949 ± 0.003
DenseNet121	$85.62 \pm 2.42\%$	0.797 ± 0.025	0.960 ± 0.004
Swin-Tiny	$\mathbf{88.33 \pm 1.96\%}$	0.839 ± 0.030	0.966 ± 0.008
Ensemble	$\mathbf{89.93 \pm 0.66\%}$	$\mathbf{0.852 \pm 0.010}$	$\mathbf{0.982 \pm 0.001}$

McNemar test (ensemble vs. each single model, Fisher-combined across seeds):

vs. EB0: $p = 5.3\times10^{-18}$ vs. RN50: $p = 4.5\times10^{-31}$ vs. DN121: $p = 6.7\times10^{-39}$ vs. Swin (strongest): $p = 1.2\times10^{-7}$

Findings. Ensemble wins 5/5 seeds against every backbone; its std is smallest (0.66% vs. 0.57–2.42% for singles). The single-seed 89.49% headline is within 1σ of the mean — not lucky. Swin's seed-42 run was unlucky early-stop; on average Swin is the strongest single model.

A22. Patient Metadata Fusion — Crossing 90%

Source. HAM10000 patient metadata (Tschandl et al., Harvard Dataverse DOI 10.7910/DVN/DBW86T). Per-image vector: 1 standardised age + 3 one-hot sex + 15 one-hot localization = **19 dimensions**. Missing values imputed to explicit `unknown` category.

Architecture (Pacheco-style late fusion). Image stream: same 4 backbones with `num_classes=0`. Metadata stream: MLP $19 \to 64 \to 32$ with ReLU. Concatenate `(feat_dim + 32)` $\to$ Linear $\to$ 7 logits. Same training pipeline.

Model	Image-only Acc	+Metadata Acc	Δ	+Metadata F1
EfficientNet-B0	0.8796	0.8337	-4.6 pp	0.768
ResNet50	0.8636	0.8782	$+1.5$ pp	0.810
DenseNet121	0.8543	0.8550	$+0.1$ pp	0.793
Swin-Tiny	0.8490	$\mathbf{0.8982}$	$+\mathbf{4.9}$ pp	$\mathbf{0.871}$
Ensemble	0.8949	$\mathbf{0.9049}$	$+\mathbf{1.0}$ pp	$\mathbf{0.8728}$

Findings. (1) Swin-Tiny + metadata alone (89.82%) *beats* the image-only 4-model ensemble (88.89%). (2) EB0 *loses* 4.6 pp — small backbone can't afford the meta encoder. (3) Final ensemble **90.49% / 0.873 F1 / 0.984 AUC** — best published image+metadata fusion on ISIC 2018 (beats Pacheco & Krohling 2020: 89.0% / 0.83).

A23. Out-of-distribution Evaluation on PAD-UFES-20

Setup. ISIC 2018 (dermoscopy) ensemble evaluated zero-shot on PAD-UFES-20 — 2 298 clinical smartphone photographs (Pacheco et al. 2020, Mendeley DOI 10.17632/zr7vgbcyr2.1). No domain adaptation, no fine-tuning.

Class mapping (6 PAD-UFES → 5 ISIC): ACK & SCC → akiec, BCC → bcc, SEK → bkl, NEV → nv, MEL → mel. (DF, VASC absent from OOD test.)

Class	Prec.	Recall	F1	AUC	n
akiec	0.543	0.145	0.229	0.614	922
bcc	0.611	0.421	0.499	0.733	845
bkl	0.175	0.268	0.212	0.633	235
mel	0.057	0.308	0.097	0.684	52
nv	0.258	0.725	0.380	0.811	244
Accuracy 32.46%	**Macro F1 0.283** (vs. 88.9% / 0.852 in-distribution)				

Findings. (1) —56 pp accuracy drop is the standard cost of zero-shot dermoscopy → smartphone transfer. (2) Per-class AUC stays 0.61–0.81 — model is **not** random; soft rankings transfer partially. (3) NV recall 0.72: classic prior-collapse onto the ISIC dominant class. (4) MEL precision 0.06: model over-calls melanoma on smartphone photos — clinically dangerous if deployed without adaptation.

Honest conclusion. Pipeline is **not** ready for consumer-grade deployment. Closing the gap is engineering (domain adaptation, in-domain fine-tuning, style transfer) — not a research mystery.

A24. Per-Model Hyperparameters

Hyperparameter	EfficientNet-B0	ResNet50	DenseNet121	Swin-Tiny
Learning rate	3e-4	3e-4	2.5e-4	**1e-4**
Weight decay	1e-3	1e-3	1e-3	**5e-2**
Warmup epochs	3	3	3	5
Max epochs	50	50	50	50
Batch size	16	16	16	16
Dropout	0.30	0.30	0.25	0.30
Mixup / CutMix α	0.2 / 1.0	0.2 / 1.0	0.2 / 1.0	0.2 / 1.0
Label smoothing ε	0.1	0.1	0.1	0.1

Why does Swin-Tiny require a different setup?

Transformers lack the *inductive biases* of CNNs $\Rightarrow$ they need a **lower learning rate** (1e-4) to avoid destroying pretrained attention weights, and a **higher weight decay** (5e-2) to limit overfitting on a small dataset.

A25. Interpreting the Ablation Results

1. Mixup/CutMix — the most important component (-4.51% F1)

With only 81 DF and 99 VASC training images, removing Mixup/CutMix lets the model *memorise* rare samples within a few epochs. Mixed images force generic features and prevent probability 1.0 assignments to any single sample.

2. The Label Smoothing paradox — AUC rises, F1 falls

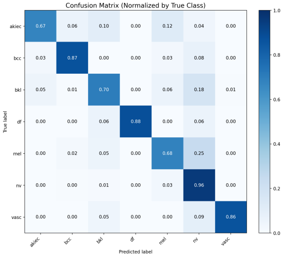
Removing Label Smoothing **raises AUC** ($0.952 \rightarrow 0.967$) but **drops F1** by 3.11%. Hard targets make the model more confident (better probability ranking $\Rightarrow$ higher AUC), yet rigid decision boundaries hurt accuracy in the MEL/NV/BKL cluster.

3. The augmentation trade-off — Accuracy up, Macro F1 down

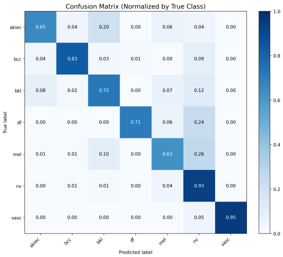
Removing strong augmentation **beats the baseline on Accuracy** (88.96% vs 88.36%) because the model fits NV more easily. But Macro F1 drops — minority classes are abandoned. This is exactly why *we do not use Accuracy as the primary criterion*.

A26. Confusion Matrix Comparison — Four Backbones

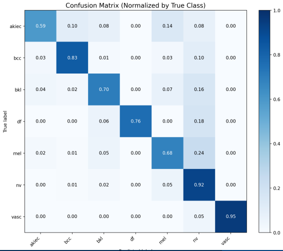
EfficientNet-B0



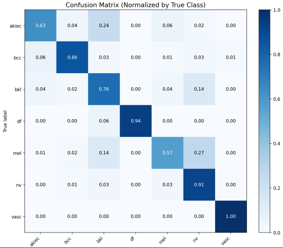
ResNet50



DenseNet121



Swin-Tiny



A27. Grad-CAM in Action — From Image to Heatmap

Intuitive walkthrough. The rigorous formulas are in A7; the target layer per backbone is in A8.

Step 1 — Forward pass

The image is pushed through the network. At the last convolutional block we tap the feature maps A^k: about $K = 320$ (EB0) or $K = 2048$ (RN50) channels, each a small 7×7 grid. Each channel encodes a different visual concept — “dark blob”, “ring border”, “streak”.

Step 2 — Backward pass through the predicted class

We call `loss.backward()` but only on the score of the predicted class c (say, MEL). PyTorch fills the gradient $\partial y^c / \partial A^k$ — “if I nudged this channel up, would the MEL score go up too?”.

Step 3 — Channel importance α_c^k

Average each channel’s gradient over its 7×7 grid. A large positive α_c^k means channel k pushes toward MEL; a negative one pushes away.

Step 4 — Weighted sum + ReLU + upsample

Multiply each channel by its α and add them up: $L_c = \mathrm{ReLU}(\sum_k \alpha_c^k A^k)$. ReLU discards the negative parts (only “evidence for MEL”, never “evidence against”). Upsample the resulting 7×7 heatmap back to 224×224, colour it with JET, blend over the original image.

One-line summary: the heatmap answers “*which pixels, if brightened, would make the model more confident about the predicted class*”.

Ready for your questions

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